

LECTURE 03

DISTANCE FORMULA

The distance between any two points $P_1 (x_1, y_1)$ and $P_2 (x_2, y_2)$ in a coordinate plane is given by

$$d(P_1, P_2) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}.$$

EXAMPLE 01

Prove that the triangle with vertices $A(-1, -3)$, $B(6, 1)$, and $C(2, -5)$ is a right triangle and find its area.

Solution: By the distance formula,

$$d(A, B) = \sqrt{(6 - (-1))^2 + (1 - (-3))^2} = \sqrt{49 + 16} = \sqrt{65}$$

$$d(B, C) = \sqrt{(2 - 6)^2 + (-5 - 1)^2} = \sqrt{16 + 36} = \sqrt{52}$$

$$d(A, C) = \sqrt{(2 - (-1))^2 + (-5 - (-3))^2} = \sqrt{9 + 4} = \sqrt{13}$$

Hence $[d(A, B)]^2 = [d(B, C)]^2 + [d(A, C)]^2$; that is, the triangle is a right triangle with hypotenuse AB . The area is $\frac{1}{2}\sqrt{52}\sqrt{13} = 13$ square units.

MIDPOINT FORMULA

The midpoint of the line segment from $P_1 (x_1, y_1)$ to $P_2 (x_2, y_2)$ is

$$\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Example 2:

Find the midpoint M of the line segment from $P_1 (-2, 3)$ to $P_2 (4, -2)$. Plot the points P_1 , P_2 , M and verify that $d(P_1, M) = d(P_2, M)$.

Solution:

Applying the midpoint formula, the coordinates of M are $\left(\frac{-2+4}{2}, \frac{3+(-2)}{2} \right)$ or $\left(1, \frac{1}{2} \right)$.

CONT...

The three points P_1 , P_2 , and M are plotted in Figure 1.11. Distance formula we obtain

$$d(P_1, M) = \sqrt{(-2 - 1)^2 + (3 - \frac{1}{2})^2} = \sqrt{9 + \frac{25}{4}} = \sqrt{61}/2$$

$$d(P_2, M) = \sqrt{(4 - 1)^2 + (-2 - \frac{1}{2})^2} = \sqrt{9 + \frac{25}{4}} = \sqrt{61}/2$$

Hence $d(P_1, M) = d(P_2, M)$.

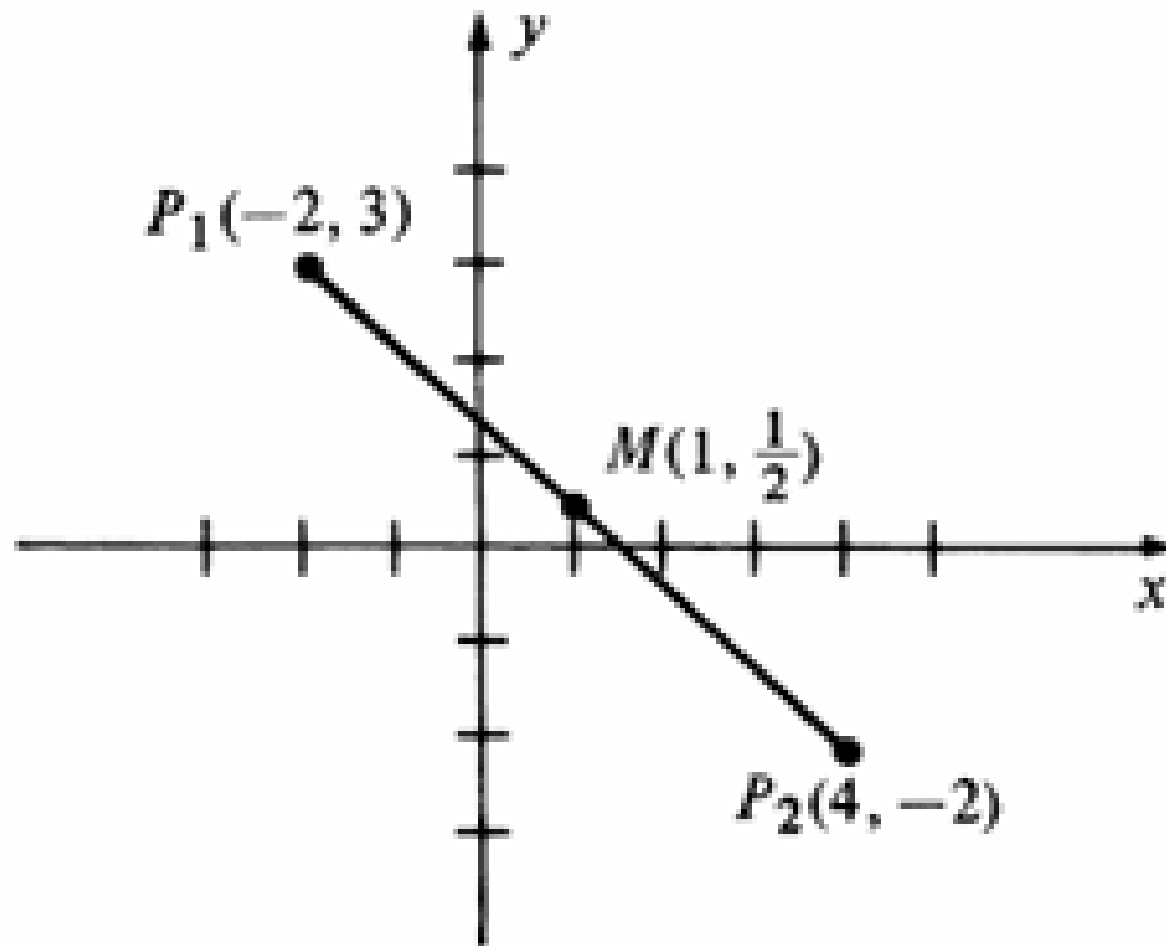


FIGURE 1.11

EXAMPLE 03

Sketch the graph of $W = \{(x, y): |x| \leq 2, |y| \leq 1\}$.

The indicated inequalities are equivalent to $-2 \leq x \leq 2$ and $-1 \leq y \leq 1$.

Hence the graph of W consists of all points within and on the boundary of the rectangular region shown in Figure 1.12.

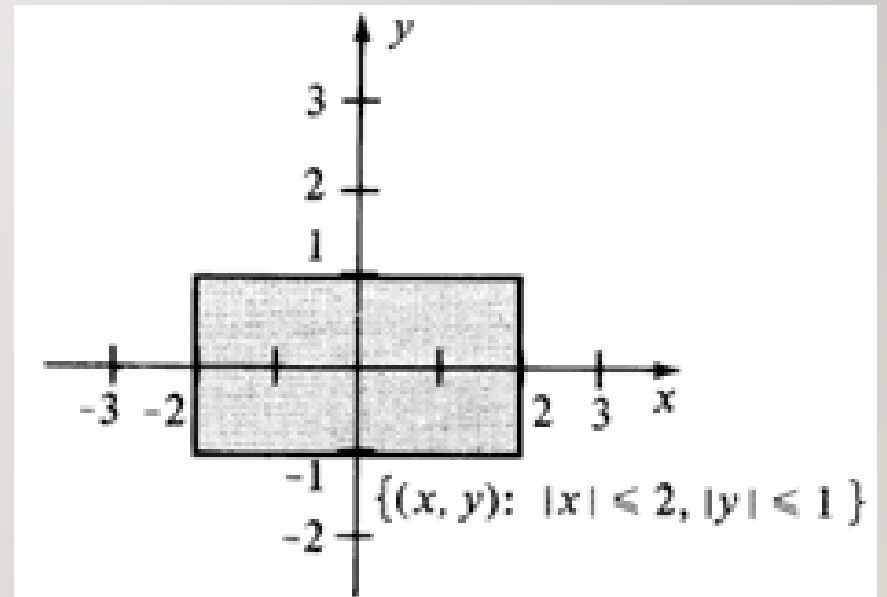


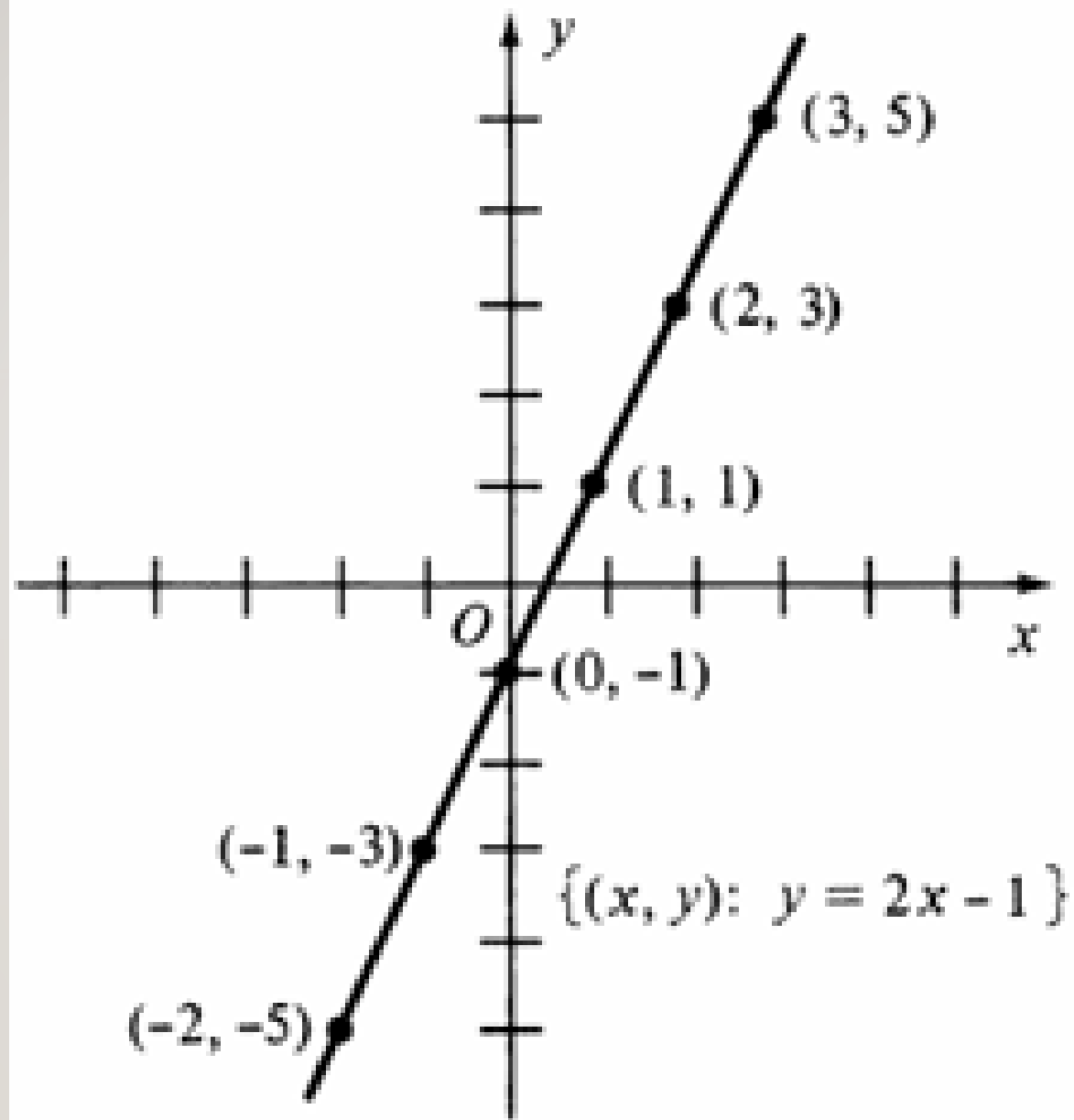
FIGURE 1.12

EXAMPLE 04

Sketch the graph of $W = \{(x, y): y = 2x - 1\}$.

We begin by finding points with the coordinates of the form (x, y) where the ordered pair (x, y) is in W . It is convenient to list these coordinates in the following tabular form, where for each real number x the corresponding value for y is $2x - 1$.

x	-2	-1	0	1	2	3
y	-5	-3	-1	1	3	5

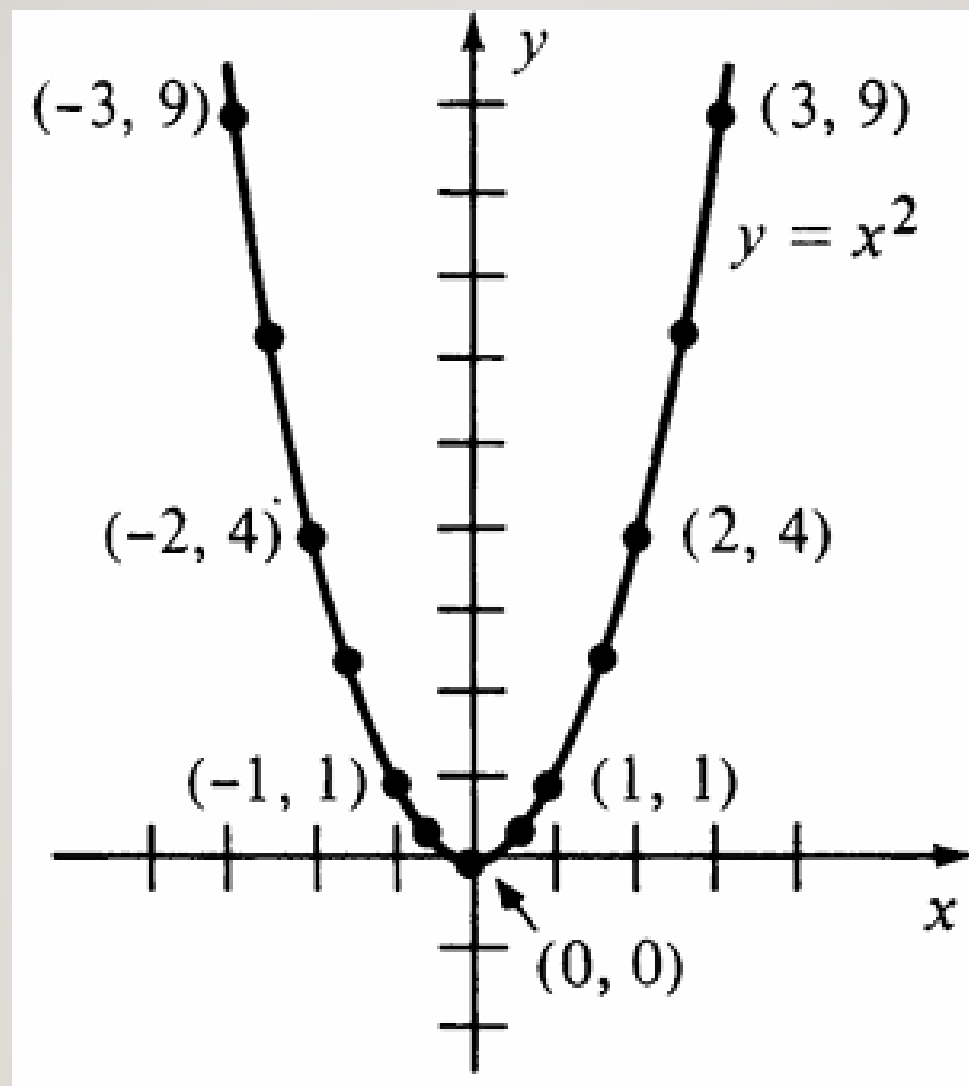


EXAMPLE 05

Sketch the graph of the equation $y = x^2$.

To obtain the graph, it is necessary to plot more points. Increasing successive x-coordinates by $\frac{1}{2}$, we obtain the following table.

x	-3	$-\frac{5}{2}$	-2	$-\frac{3}{2}$	-1	$-\frac{1}{2}$	0	$\frac{1}{2}$	1	$\frac{3}{2}$	2	$\frac{5}{2}$	3
y	9	$\frac{25}{4}$	4	$\frac{9}{4}$	1	$\frac{1}{4}$	0	$\frac{1}{4}$	1	$\frac{9}{4}$	4	$\frac{25}{4}$	9



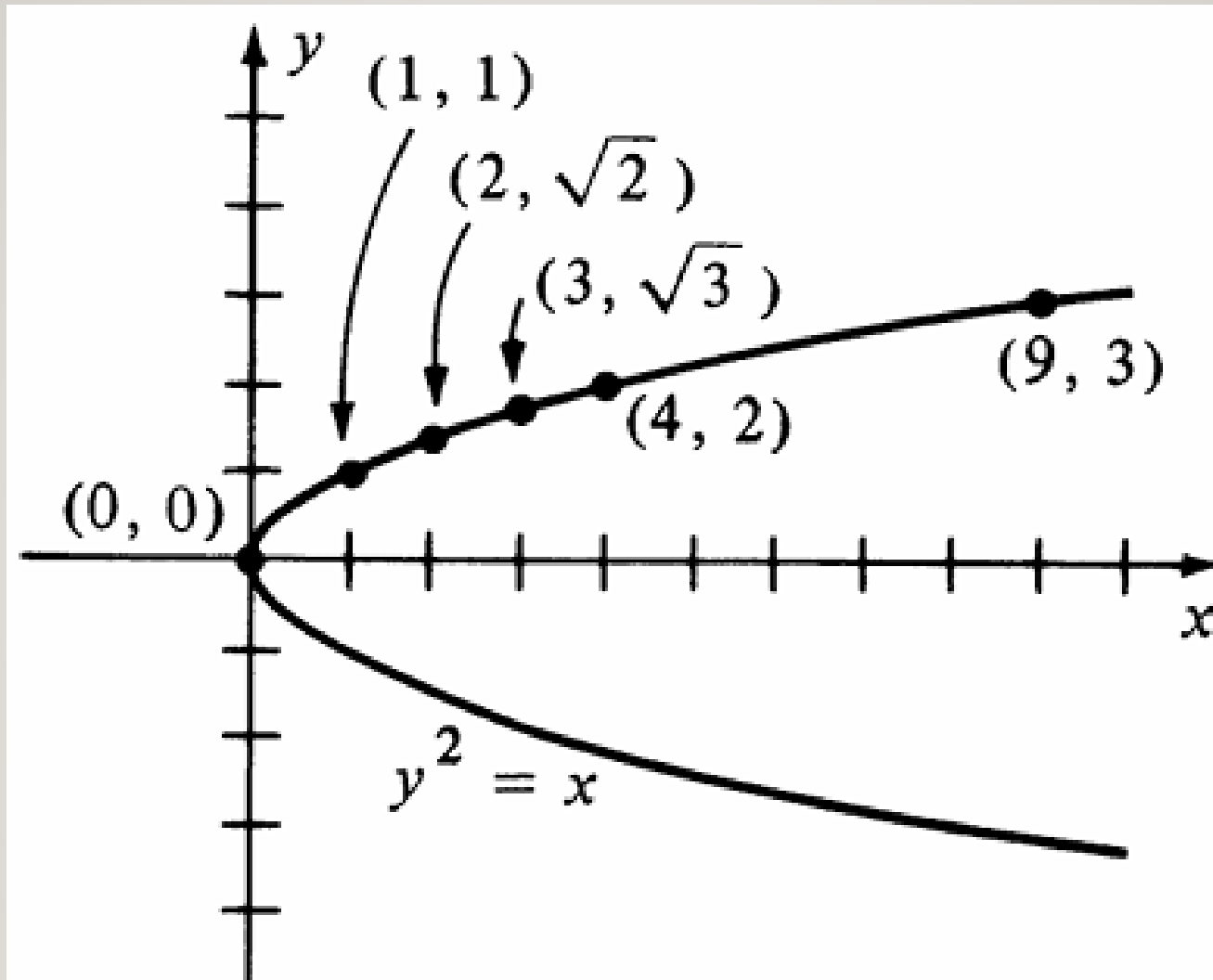
EXAMPLE 06

Sketch the graph of the equation $y^2 = x$.

Since substitution of $-y$ for y does not change the equation, the graph is symmetric with respect to the x -axis. It is sufficient, therefore, to plot points with nonnegative y -coordinates and then reflect through the x -axis. Since $y^2 = x$, the y -coordinates of points above the x -axis are given by $y = \sqrt{x}$. Coordinates of some points on the graph are tabulated below.

The graph is a parabola that open to the right, with its vertex at the origin.

x	0	1	2	3	4	9
y	0	1	$\sqrt{2}$	$\sqrt{3}$	2	3



EXAMPLE 07

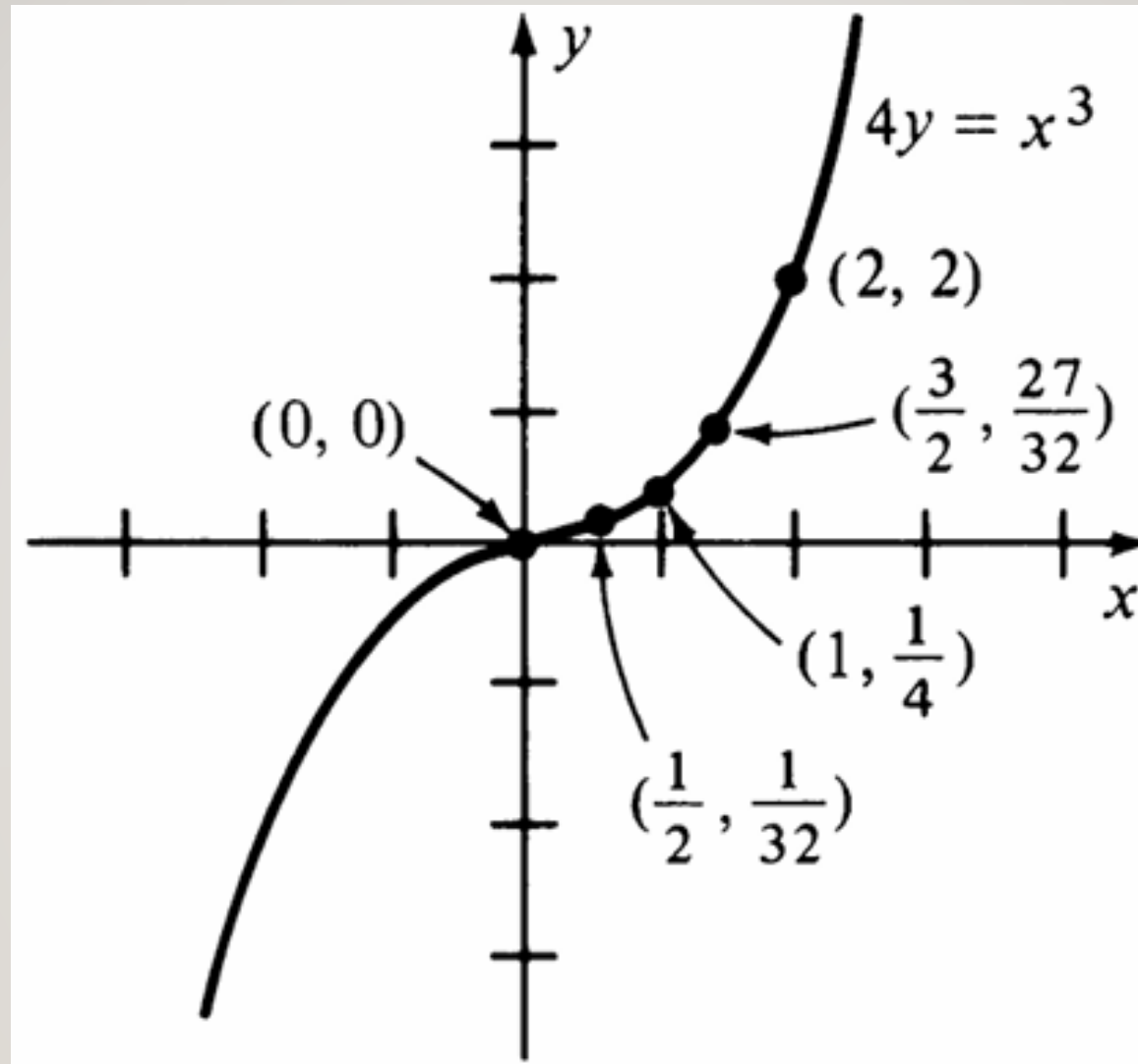
Sketch the graph of the equation $4y = x^3$.

If we substitute $-x$ for x and $-y$ for y , then

$$4(-y) = (-x)^3 \quad \text{or} \quad -4y = -x^3.$$

Multiplying both sides by -1 , we see that the last equation has the same solutions as the given equation $4y = x^3$. Hence, from symmetry test the graph is symmetric with respect to the origin. The following table lists some points on the graph.

x	0	$\frac{1}{2}$	1	$\frac{3}{2}$	2	$\frac{5}{2}$
y	0	$\frac{1}{32}$	$\frac{1}{4}$	$\frac{27}{32}$	2	$\frac{125}{32}$



CIRCLE

If $C(h, k)$ is a point in a coordinate plane, then a circle with center C and radius $r > 0$ may be defined as the collection of all points in the plane that are r units from C .

A point $P(x, y)$ is on the circle if and only if $d(C, P) = r$ or, by the distance formula, if and only if

$$\sqrt{(x - h)^2 + (y - k)^2} = r$$

The equivalent equation

$$(x - h)^2 + (y - k)^2 = r^2$$

is an equation of a circle of radius r and center $C(h, k)$.

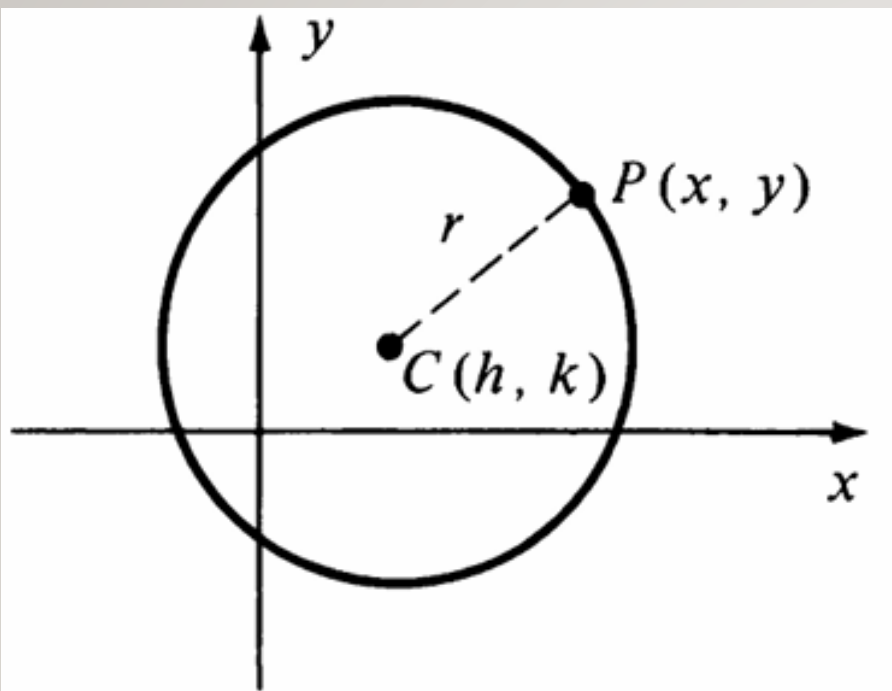
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If $h = 0$ and $k = 0$, this equation reduces to

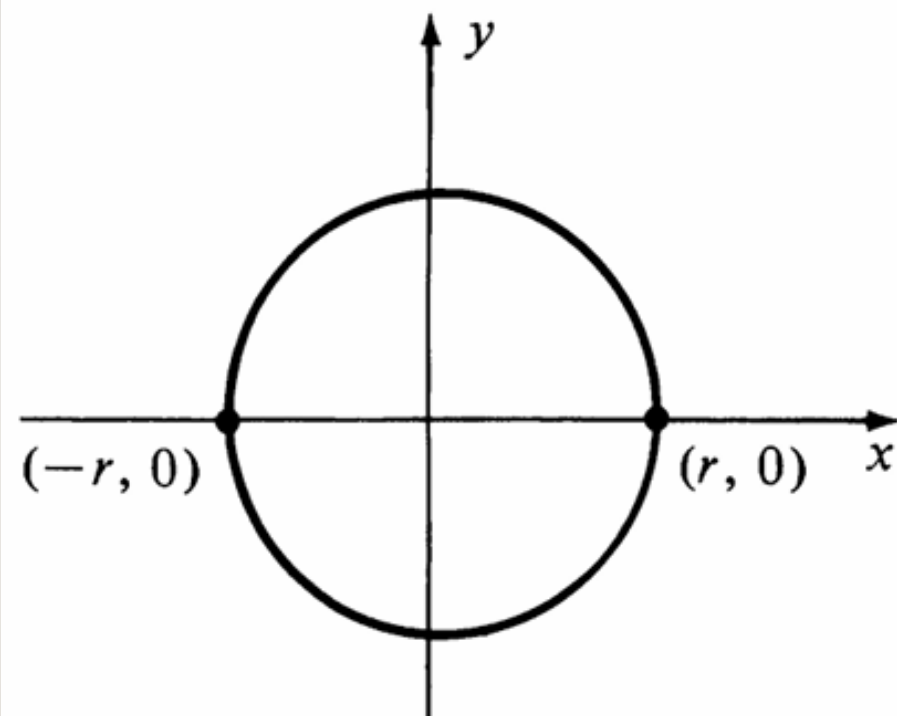
$$x^2 + y^2 = r^2,$$

Which is an equation of circle of radius r with center at the origin.

If $r = 1$, the graph is called a **unit circle**.



(i) $(x - h)^2 + (y - k)^2 = r^2$



(ii) $x^2 + y^2 = r^2$

EXAMPLE 08

Find an equation of the circle with center $C(-2,3)$ which passes through the point $D(4, 5)$.

Solution:

Since D is on the circle, the radius r is $d(C, D)$. By the distance formula

$$r = \sqrt{(-2 - 4)^2 + (3 - 5)^2} = \sqrt{36 + 4} = \sqrt{40}$$

Using with $h = -2$ and $k = 3$, we obtain

$$\begin{aligned}(x + 2)^2 + (y - 3)^2 &= 40, \\ x^2 + y^2 + 4x - 6y - 27 &= 0.\end{aligned}$$

In Exercises 1–6, find (a) the distance $d(A, B)$ between the points A and B , and (b) the midpoint of the segment AB .

1 $A(6, -2), B(2, 1)$

2 $A(-4, -1), B(2, 3)$

3 $A(0, -7), B(-1, -2)$

4 $A(4, 5), B(4, -4)$

5 $A(-3, -2), B(-8, -2)$

6 $A(11, -7), B(-9, 0)$

In Exercises 7 and 8 prove that the triangle with vertices A , B , and C is a right triangle and find its area.

7 $A(-3, 4), B(2, -1), C(9, 6)$

8 $A(7, 2), B(-4, 0), C(4, 6)$

In Exercises 15–20 sketch the graph of the set W .

15 $W = \{(x, y): x = 4\}$

16 $W = \{(x, y): y = -3\}$

17 $W = \{(x, y): xy < 0\}$

18 $W = \{(x, y): xy = 0\}$

19 $W = \{(x, y): |x| < 2, |y| > 1\}$

20 $W = \{(x, y): |x| > 1, |y| \leq 2\}$

In Exercises 39–46 find an equation of a circle satisfying the stated conditions.

39 Center $C(3, -2)$, radius 4

40 Center $C(-5, 2)$, radius 5

41 Center at the origin, passing through $P(-3, 5)$

42 Center $C(-4, 6)$, passing through $P(1, 2)$

43 Center $C(-4, 2)$, tangent to the x -axis

44 Center $C(3, -5)$, tangent to the y -axis

45 Endpoints of a diameter $A(4, -3)$ and $B(-2, 7)$

46 Tangent to both axes, center in the first quadrant, radius 2